

Day 3: Consumer Credit Portfolio Risk

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1 Part 1: Data Cleaning, EDA & Data Integrity

1.1 Overview

The raw loan tape contained 308,222 records and 12 fields. After a documented cleaning pipeline, 277,867 loans (90.2%) remain in the analysis-ready dataset. The pipeline distinguishes between defects that are **corrected**, **flagged and retained**, and **removed as structurally unusable**, since these carry different implications for downstream survival and copula modelling.

1.2 Data-Quality Log

Defect	Affected	% Raw	Treatment
Duplicate <code>loan_id</code>	6,164	2.00%	Kept first occurrence; later duplicates removed
Invalid/unrecognised region (post-mapping)	0	0.00%	Twelve spelling/typo variants mapped to four canonical regions; none remained unmapped
Invalid amortization type (XYZ)	1,202	0.39%	Removed; EAD cannot be reconstructed
Missing amortization type	1,191	0.39%	Removed; EAD cannot be reconstructed
Other unrecognised amortization spellings	2,393	0.78%	Removed
Invalid FICO score (outside 300–850)	6,028	1.96%	Set to missing; loan retained and flagged
Invalid interest rate (≤ 0 or $> 100\%$)	3,419	1.11%	Set to missing; loan retained and flagged
Non-positive term	3,048	0.99%	Removed
Term above 120 months	2,987	0.97%	Removed
Non-positive issue amount	5,990	1.94%	Removed
Implausible issue amount ($> 12,560$)	3,082	1.00%	Systematic unit/decimal defect; capped at €12,560 and flagged
Missing/unparseable origination date	10,472	3.40%	Removed
Default date before origination	5,733	1.86%	Default date set to missing and flagged
Prepayment date before origination	4,199	1.36%	Prepayment date set to missing and flagged
Default after <code>obs_end_date</code>	795	0.26%	Treated as administratively censored
Prepayment after <code>obs_end_date</code>	750	0.24%	Treated as administratively censored
Both default and prepayment recorded	5,782	1.88%	Earlier event retained; competing-risk resolution
— default preceded prepayment	2,516	0.82%	Prepayment dropped
— prepayment preceded default	3,264	1.06%	Default dropped
— same-day (ambiguous)	2	0.00%	Both dates dropped; loan treated as censored
Maturity inconsistent with <code>origination_date + term</code>	5,600	1.82%	Replaced with reconstructed contractual maturity
Structurally unusable (net removals)	24,191	7.85%	Removed; age, schedule, or EAD not reliably reconstructable

Table 1: Part 1 data-quality log: defects detected and treatment applied.

Net effect: 308,222 $\rightarrow$ 277,867 loans (30,355 removed, 9.85%).

1.3 Descriptive Profile (Cleaned Tape)

- **Portfolio size:** 277,867 loans; total issue amount €81.6bn; total outstanding balance at `obs_end_date` €4.32bn.
- **Term:** trimodal at 12 / 36 / 60 months (contractual buckets only — no interpolated terms).
- **FICO:** approximately uniform across 300–850 after cleaning.
- **Interest rate:** right-skewed, mode ~10–12%, with a thin tail up to 75% (retained — plausible for subprime unsecured credit, but flagged for sensitivity testing rather than assumed benign).
- **Region split:** Southeast 72,496 / Southwest 71,804 / Pacific 71,073 / Northeast 62,494.
- **Amortization split:** effectively even across French / Italian / Bullet (~92,600 each).

1.4 Censoring, Default & Prepayment Rates

Overall composition:

Event	Count	%
Censored	222,144	79.95%
Default	28,476	10.25%
Prepayment	27,247	9.81%

By region:

Region	Default Rate	Prepayment Rate
Northeast	18.14%	9.38%
Pacific	8.84%	9.82%
Southwest	8.68%	9.76%
Southeast	6.38%	10.21%

By FICO band:

Band	Default Rate	Prepayment Rate
<580	14.61%	9.66%
580–669	5.76%	9.89%
670–739	5.67%	10.02%
740–799	5.72%	9.93%
800+	5.69%	9.90%

The Region $\times$ FICO heatmap shows the Northeast < 580 cell at 25.5% default, roughly 2.5 $\times$ the next-highest cell; indicating the regional effect is concentrated in the lowest credit tier rather than uniform across FICO bands. This interaction is the working hypothesis carried into segmentation (Part ??).

With ~80% censoring, the *tail* of any fitted marginal, and especially any fitted copula tail-dependence coefficient, will carry material sampling uncertainty; flagged here for the estimation-risk discussion in Part 6.

1.5 Amortization & EAD

Outstanding balance reconstructed per loan from `amortization_type`, `term`, `interest_rate`, `issue_amount`, assuming monthly payment frequency and a nominal annual rate divided by 12:

- **French:** constant annuity, principal component compounding.
- **Italian:** equal principal payments.

- **Bullet:** full principal outstanding until maturity (interest-only).

Amortization	Total Balance at obs_end_date	Mean EAD at Default	Median EAD at Default
Bullet	€1.96bn	€217,475	€6,108
French	€1.31bn	€143,738	€2,663
Italian	€1.06bn	€148,034	€2,368

Bullet loans hold materially higher balances at default than French or Italian loans, as expected structurally (no principal amortization prior to maturity). This supports the working hypothesis that amortization type will matter for loss severity independent of default frequency.

Validation checks (no violations found): no default/prepayment before origination, no event after obs_end_date, no negative EAD or balance, no EAD or balance exceeding issue amount.

1.6 Structural Data-Quality & Survivorship Risks

Beyond the mechanical defects above, at least three structural issues could contaminate inference, with their direction of bias:

1. **Administrative censoring (~80% of the book).** If censored loans were treated as “survived,” realized default rates would be biased **downward**, understating true credit risk. Handled here as right-censoring (Part 2), not as non-events.
2. **Survivorship in the securitization pool.** The pool consists of loans that survived long enough to be selected for the deal; loans that defaulted very early in their life may be underrepresented relative to the full origination population. This biases observed default risk **downward** relative to the true historical book.
3. **Regional/system reporting inconsistency.** Twelve distinct raw spellings for four regions suggest separate servicing systems or entry conventions. To the extent data quality itself varies by region, apparent regional risk differentials may be **inflated or deflated** by recording artifacts rather than true economic risk — a caveat for the regional dependence story in Part 3.

1.7 Implausible Issue Amounts

Some loan amounts in the raw data are clearly much larger than would be expected for unsecured personal loans. The typical loan amount is €6,830, but some values are extremely large. There are also no loans between €12,560 and €5,006,130, which suggests that these large values are due to a data-entry error rather than genuine loan amounts.

About 3,017 loans (0.98% of the raw data) are affected. The pattern is consistent with some amounts being recorded incorrectly, for example because of an extra digit or misplaced decimal point.

Treatment: The affected loans were kept because the other information in their records was still usable. Their issue amounts were capped at €12,560, which is the point where the unusual values begin. The original amounts were retained in `issue_amount_original`, and the affected loans were marked using `issue_amount_extreme_flag`. This prevents the incorrect values from distorting the analysis while keeping the useful loan records.

1.8 Working Hypotheses Carried Forward

1. FICO score is the strongest single default driver (<580 defaults at $\sim 2.5\times$ the rate of every higher band).
2. Regional effects are material and concentrated in the low-FICO tier (Northeast $\times$ <580 far exceeds any other cell).
3. Amortization type drives loss *severity* (EAD at default) more than default *frequency*; relevant to how LGD/EAD is modelled in Part 5.

4. A systematic issue-amount entry defect (1.00% of loans, capped) was driving apparent tail concentration mechanically in the first simulation pass; Parts 5/6 were re-run to confirm concentration findings reflect genuine segment risk rather than this artifact.
5. The strong regional clustering in defaults is consistent with a shared systematic factor rather than independent obligor risk; tested formally in Part 3.

2 Part 2: Marginal Estimation

2.1 Time Scale and Competing-Risk Treatment

Loan age is measured in months from `origination_date`. Default is the main event of interest, while prepayment and administrative censoring are alternative ways in which a loan can leave the analysis.

Two approaches were considered. The cause-specific approach was used for comparing the default-time models and for obtaining the parameters used in Part 3. The Aalen–Johansen method was used as a cross-check that explicitly accounts for prepayment as a competing event.

The censoring definition was also corrected so that loans are no longer treated as active beyond their contractual maturity. Censored loans are now followed only until the earlier of the observation end date or maturity date.

2.2 Family Comparison (Final)

Three distributions were compared with the Kaplan–Meier estimate. The results are shown below.

Model	Log-Lik.	AIC	Akaike Weight	RMSE vs. KM	AIC Rank
Weibull	−190,726.74	381,457.49	1	0.1663	1
Log-Logistic	−190,884.30	381,772.60	0	0.1708	2
Log-Normal	−191,660.04	383,324.08	0	0.1827	3

Table 2: Comparison of candidate marginal default-time models.

The **Weibull model was selected**, with an Akaike weight of approximately 1.00, indicating that it provided the strongest fit among the three models.

2.3 Final Model Parameters

The fitted parameters for all three models are retained below for reference.

Parameter	Weibull	Log-Logistic	Log-Normal
Shape (ρ)	1.2353	—	—
Scale (λ)	207.90	—	—
α	—	185.63	—
β	—	1.2800	—
μ	—	—	5.6868
σ	—	—	1.7354

Table 3: Fitted parameters for the candidate marginal default-time models.

For the selected Weibull model, the shape parameter is $\rho = 1.235 > 1$, indicating that the likelihood of default increases as the loan gets older. This pattern is also visible in the Aalen–Johansen cumulative incidence curve, particularly toward the later months of the loan term (Figure 1).

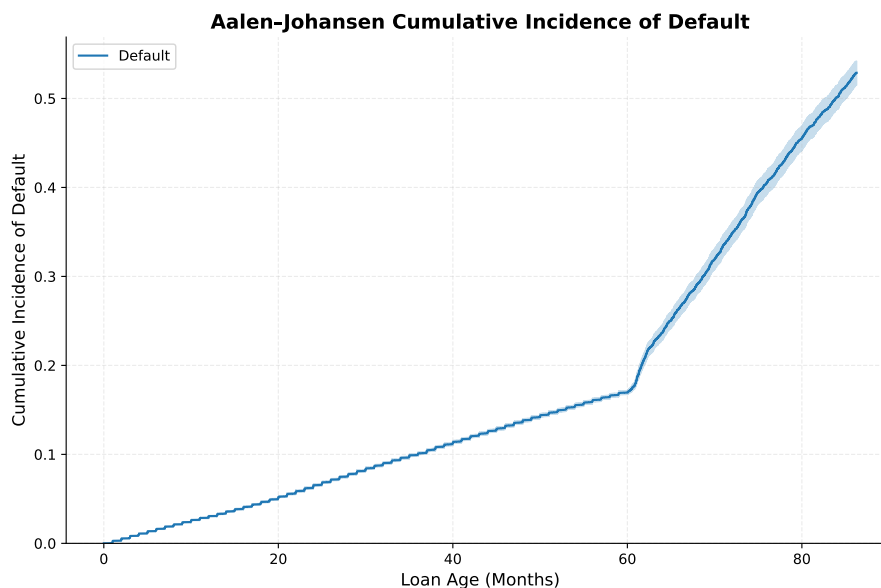


Figure 1: Aalen–Johansen cumulative incidence of default by loan age.

Grouping	Params	Log-Lik.	AIC	BIC	Δ AIC
Credit score (5 groups)	10	−183,690.86	367,401.71	367,506.86	0.0
Region (4 groups)	8	−186,104.27	372,224.54	372,308.82	4,822.8
Loan length (3 groups)	6	−189,844.40	379,700.80	379,764.01	12,299.1
No grouping	2	−190,726.74	381,457.49	381,478.56	14,055.8
Repayment type (3 groups)	6	−190,725.74	381,463.48	381,526.69	14,061.8

Table 4: Comparison of different ways of grouping loans. Lower AIC and BIC indicate a better fit.

2.4 Which Loans Behave Alike? Grouping by Credit Score

We compared four ways of grouping loans: credit score, region, loan length, and repayment type. Credit score gave the best fit by a wide margin, while the other groupings added much less information.

The results show that **credit score is the most useful way to group the loans**. In particular, loans with scores below 580 have a much higher default rate than the other groups.

Credit Score Band	N	Default Rate	Shape	Scale
800+	25,053	5.69%	1.4436	256.31
740–799	29,840	5.72%	1.3946	272.43
670–739	34,254	5.67%	1.4329	260.44
580–669	44,192	5.76%	1.3869	272.23
Below 580	138,986	14.61%	1.1953	157.81

Table 5: Default behaviour across credit score groups.

Loans with scores below 580 default much more often than the other groups, confirming that credit score is the strongest factor for distinguishing default behaviour.

2.5 Follow-Up Check: Effect of Loan Length

The models fit the data well up to around month 60 and then begin to diverge. (see Figure 3. A check of loan length helps explain this pattern. Longer loans have higher default rates: 12-month loans defaulted

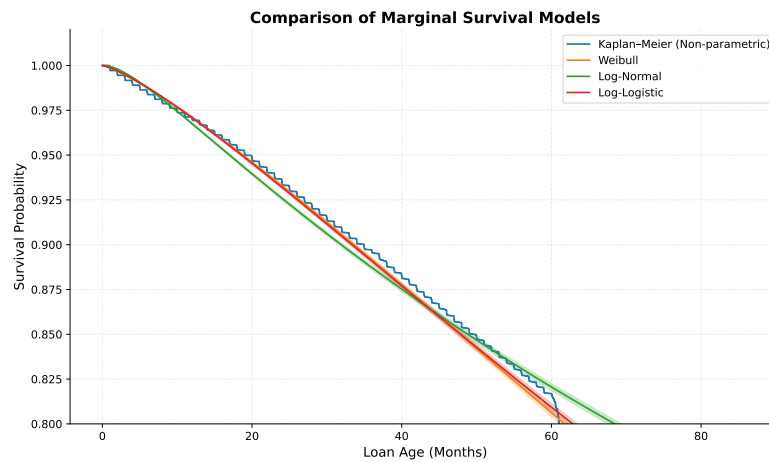


Figure 2: Model estimates compared with the Kaplan–Meier estimate of loans remaining active over time.

at 4.1%, compared with 10.3% for 36-month loans and 16.2% for 60-month loans.

After month 60, most shorter loans have already matured, leaving mainly the longer and riskier loans. Thus, the change after month 60 is largely explained by the changing mix of loans over time rather than a problem with the model.

Loan length is therefore not used as the main grouping variable; **credit score provides a much stronger distinction between loans.**

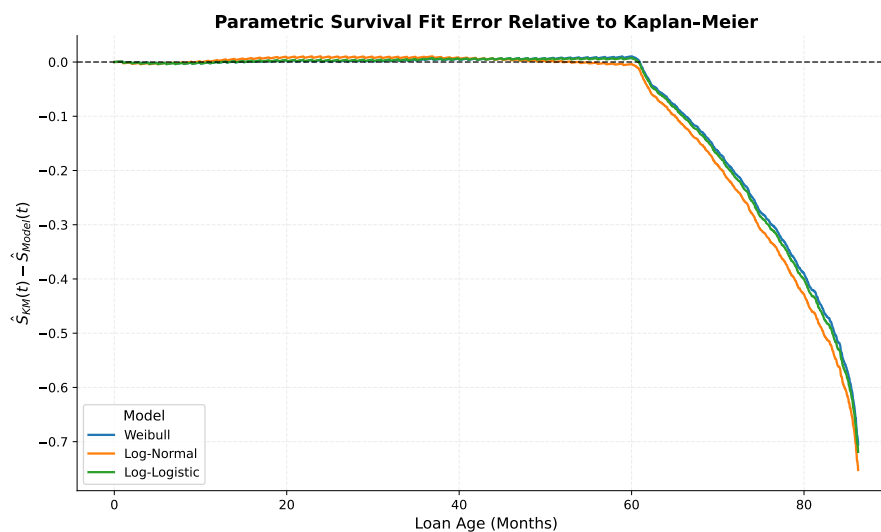


Figure 3: Gap between each model and the plain Kaplan–Meier estimate. All three models agree well up to month 60, then diverge.

3 Part 3: Copula Selection and Dependence

3.1 Dependence Between Loans

Loans do not behave independently: when default rates rise in one group, they tend to rise in other groups at the same time. This suggests the presence of a shared factor affecting many loans at once.

Because around 80% of loans are censored, dependence was estimated using annual default rates for each FICO band and region rather than individual loan records. Kendall's τ was used to measure how

closely groups move together. This is an aggregate measure of dependence and is therefore treated as an approximation rather than a direct loan-level measure.

3.2 Dependence by Credit Score and Region

The average dependence between FICO bands was $\tau = 0.379$, corresponding to a correlation of $\rho = 0.5602$. All pairwise relationships were positive, including between the highest- and lowest-score groups, suggesting that a common factor affects loans across different credit-quality groups.

	580–669	670–739	740–799	800+	<580
580–669	1.000	0.264	0.396	0.347	0.410
670–739	0.264	1.000	0.458	0.288	0.329
740–799	0.396	0.458	1.000	0.359	0.530
800+	0.347	0.288	0.359	1.000	0.405
<580	0.410	0.329	0.530	0.405	1.000

Table 6: Dependence between FICO bands measured by Kendall’s τ .

Regional default rates also move together over time, even though the regions have different overall default levels.

	Northeast	Pacific	Southeast	Southwest
Northeast	1.000	0.408	0.267	0.356
Pacific	0.408	1.000	0.313	0.520
Southeast	0.267	0.313	1.000	0.431
Southwest	0.356	0.520	0.431	1.000

Table 7: Dependence between regions measured by Kendall’s τ .

3.3 Testing for an Additional FICO-Specific Effect

We tested whether loans within the same FICO band move together more strongly than loans from different FICO bands.

Measure	Value
Same FICO band	0.2712
Different FICO bands	0.2547
Difference	0.0165
Test p -value	0.403

Table 8: Comparison of dependence within and across FICO bands.

The difference is small and not statistically significant. Therefore, there is **no clear evidence of a separate FICO-specific dependence factor**. A single common factor is therefore used rather than adding a second factor.

3.4 Final Copula Selection

The Student- t copula was selected because it can represent both the observed dependence and the tendency for extreme outcomes to occur together. The fitted parameters are $\rho = 0.5602$ and $\nu = 2.74$.

Final choice: a **One-Factor Student- t copula** with $\rho = 0.5602$ and $\nu = 2.74$, combined with the FICO-band Weibull models from Part 2. This provides a simple way to represent the common movements and extreme co-movements observed in the data without adding an unsupported second factor.

Structure	Factors	ν	ρ	λ_U	Decision
Independence	0	—	—	0.000	Rejected; loans clearly move together
One-Factor Gaussian	1	—	0.560	0.000	Rejected; cannot capture joint extreme outcomes
One-Factor Student-t	1	2.74	0.560	0.367	Selected

Table 9: Comparison of dependence structures.

4 Part 4: Diversification Analysis

Critical question: Does regional diversification reduce tail risk?

Regional default rates differ substantially, from 18.1% in the Northeast to 6.4% in the Southeast. However, all regional pairs show positive dependence ($\tau = 0.27$ – 0.52). Thus, regions differ in their default *levels* but still tend to experience defaults at similar times.

Conclusion: regional diversification does not provide strong protection against extreme losses because all four regions remain exposed to the same systematic factor identified in Part 3. Tail-risk capital should therefore be based on the dependent Student- t model rather than an independence assumption.

5 Part 5: Monte Carlo Portfolio Loss & Risk Measures

5.1 Simulation Design

The final simulation uses a FICO-stratified sample of 4,900 loans ($\sim 1.76\%$ of the cleaned portfolio), a 36-month horizon, and 10,000 Monte Carlo paths. FICO-band Weibull marginals from Part 2 are combined with the one-factor Student- t copula from Part 3 ($\rho = 0.5602$, $\nu = 2.74$).

EAD is recalculated at each simulated default time, with LGD fixed at 45%. Missing interest rates are replaced by the corresponding FICO-band median. Issue amounts remain capped at €12,560 as specified in Part 1.

Important: all results below are at the 4,900-loan sample scale and are not full-portfolio estimates.

5.2 Risk Measures

Structure	EL	VaR _{95%}	VaR _{99%}	VaR _{99.9%}	ES _{95%}	ES _{99%}	ES _{99.9%}	EC _{99.9%}
Student- t (dep.)	€896,954	€4.49M	€8.61M	€12.86M	€7.01M	€10.73M	€13.76M	€11.97M
Independence	€895,871	€0.98M	€1.01M	€1.05M	€1.00M	€1.03M	€1.07M	€0.15M

The expected loss is almost unchanged by the dependence assumption, but the tail risk is much higher under the Student- t model. At the 99.9% level, economic capital is €11.97M compared with only €0.15M under independence.

Monte Carlo confidence intervals: the 500-bootstrap 99.9% VaR intervals are [€12.29M, €13.72M] for the Student- t model and [€1.04M, €1.06M] under independence.

The Euler check gives a direct VaR_{99.9%} of €12.86M versus €13.76M from the sum of Component VaR, a 6.99% difference consistent with Monte Carlo variation at 10,000 paths.

5.3 Independence Benchmark

The difference between the two models is substantial:

$$1 - \frac{\text{VaR}_{99.9\%}(\text{independent})}{\text{VaR}_{99.9\%}(\text{dependent})} = \mathbf{91.8\%}.$$

Thus, assuming independent defaults would understate 99.9% tail capital by 91.8%, while expected loss changes very little. Economic capital is roughly 11 times higher when common dependence and heavy-tailed co-movement are included.

5.4 Scale Caveat

The simulation covers only 4,900 of the 277,867 cleaned loans. A proportional gross-up by $\times 56.7$ gives an indicative $EC_{99.9\%}$ of approximately €679M, compared with total outstanding balance of €4.32bn ($\sim 16\%$).

This is only an approximation because tail risk does not scale linearly under dependence. A full-portfolio simulation should therefore be used if computationally feasible; otherwise, the €679M figure should be clearly labelled as a proportional gross-up rather than a full-portfolio estimate.

6 Part 6: Concentration, Attribution & Critique

6.1 How Concentrated Is the Risk?

Measure	Value
Concentration by exposure	0.0003
Concentration by risk contribution	0.0003
Share of risk from the 10 riskiest loans	0.41%
Diversification ratio	1.0699

Table 10: Concentration measures after the issue-amount fix.

Before the issue-amount fix, a handful of corrupted loan records made the portfolio look far more concentrated than it really is (10 loans accounted for 97.7% of tail risk). With the cap applied, risk is spread across the book rather than driven by a few loans, as expected in a large pool of ordinary consumer loans.

6.2 Which Regions Carry the Most Risk?

Region	Risk Contribution
Northeast	€610,827
Pacific	€447,941
Southwest	€446,306
Southeast	€307,149

Table 11: Risk contribution by region.

The Northeast has both the highest default rate and strong co-movement with every other region (Part 1, Part 3), so it contributes the largest share of tail risk — about 27% of the total. No single region or loan dominates the portfolio, consistent with the diffuse concentration picture above.

6.3 Critique

Model risk. We're not fully certain about ν , the parameter controlling how strongly bad outcomes cluster together. Our estimate is 2.74, but the data supports anywhere from 2.05 to 4.79. Testing all three:

Capital moves by about $\pm 7\%$ across this range; real, but smaller than the range itself suggests. Expected loss barely moves, since ν affects whether bad things happen together, not how much is lost on average. LGD and the marginal model weren't re-tested but should matter less.

Estimation risk. Our dependence estimates rest on only ~ 34 years of annual data; thin for anything about rare events. That's why ν 's range is wide, and why the earlier finding of no extra credit-score

ν	Scenario	EL	VaR _{99%}	VaR _{99.9%}	ES _{99.9%}	EC _{99.9%}
2.05	Heaviest tail	€904,352	€8.98M	€13.66M	€14.27M	€12.75M
2.74	Base case	€896,954	€8.61M	€12.86M	€13.76M	€11.97M
4.79	Lightest tail	€914,362	€8.23M	€12.02M	€13.25M	€11.10M

Table 12: Effect of ν on capital, holding everything else fixed.

factor (Part 3) should be read as "not detected," not "doesn't exist." A stronger check would rerun the full pipeline on resampled data rather than treating each step's uncertainty in isolation.

Gap to a real use case. Left out: adjusting for the economic cycle, letting losses worsen in a downturn, realistic prepayment behavior, and regulatory requirements (Basel, IFRS 9). We also assumed a specific statistical structure that real-world data isn't guaranteed to follow.

Recommendation. Assuming independent defaults would understate capital by 92%; the dependence modeling matters. Capital should be based on the dependent model (€11.97M on the sample; ~€679M scaled to the full portfolio, pending a full run). Two flags: hold capital nearer €12.75M given the uncertainty in ν , and fix LGD next, since it's currently static and likely understates downturn losses.

Executive Summary (Investor Audience)

- **Expected loss:** €896,954 on the sample. Barely changes whether or not defaults are modeled as dependent; that only affects the worst case, not the average.
- **Capital at 99.9%:** €11.97M on the sample; ~€679M scaled to the full portfolio (pending a full run to confirm).
- **Main finding:** independent defaults would understate capital by 92%. Regions and credit bands default at different rates but tend to default at the same times; so diversification helps less than it looks.
- **Main caveat:** the clustering parameter is uncertain (~34 years of data). Its plausible range moves capital by $\pm 7\%$; real, but not enough to change the conclusion. Hold capital toward the higher end as a margin.